

Supplementary material

How much of hazard news is news? Decomposing open-source coverage of chemical accidents and CBRN threats for hazard surveillance

S1. Domestic EVENT articles in the security block (gold set)

Query	Date	Coded place	Coded object
White-powder report	2022-04-27	인천 부평구	백색가루
White-powder report	2022-06-12	대구 수성구	백색가루
White-powder report	2022-06-12	대구 수성구	백색가루
White-powder report	2022-06-13	대구 수성구	백색가루
White-powder report	2022-07-04	대구 수성구	백색가루
Anthrax terror	2022-09-28	제주시 조천읍	백색가루 우편물
Anthrax terror	2022-09-28	제주시 조천읍	탄저균 의심 우편물
Anthrax terror	2022-11-04	제주시 오라동	정체불명 우편물
Bioterrorism	2023-07-21	함안군 칠원읍	생물테러 의심 소포
Anthrax terror	2023-07-22	제주시	독극물 소포
Radiation leak	2023-07-24	영광군 한빛원전	원전 자동정지
DPRK nuclear test	2024-05-01	제주시	공중음파
Radiation leak	2024-08-07	경주시	원전 비상발전기 작동
White-powder report	2024-08-29	부산 해운대구	백색가루 우편물
DPRK bio-chem weapons	2024-09-19	경기도	쓰레기풍선
DPRK bio-chem weapons	2024-10-04	서울	오물풍선
Radiation leak	2025-04-09	영광군	원전 비상디젤발전기 기동
Radiation leak	2025-08-21	부산 기장군	원전 연기

S2. Leave-one-query-out evaluation of the proposed classifier

Trained on 29 queries, tested on the held-out query. p_E true and predicted are percentages.

Query	Block	n	EVENT	Macro- F_1	EVENT- F_1	p_E true	p_E pred
HF leak	industrial	100	40	0.73	0.84	40	43
NH ₃ leak	industrial	100	50	0.84	0.95	50	53
Cl ₂ gas leak	industrial	100	35	0.74	0.87	35	36
H ₂ SO ₄ spill	industrial	100	69	0.68	0.98	69	72
Chem. plant explosion	industrial	100	47	0.48	0.86	47	58
Hazmat release	industrial	67	6	0.58	0.91	9	7
Toluene leak	industrial	100	40	0.74	0.93	40	42
HNO ₃ leak	industrial	100	73	0.60	0.95	73	75
Toxic-gas evacuation	industrial	100	55	0.53	0.86	55	63
Chem. accident response	industrial	100	3	0.46	0.40	3	2
Chemical weapon use	security	100	11	0.46	0.24	11	6
Sarin attack	security	41	3	0.44	0.57	7	10
Bioterrorism	security	100	1	0.64	1.00	1	1
Anthrax terror	security	100	7	0.45	0.67	7	5
DPRK nuclear test	security	100	7	0.13	0.00	7	0
Radiation leak	security	100	28	0.46	0.38	28	9
Dirty bomb	security	100	4	0.34	0.60	4	6
White-powder report	security	59	7	0.34	0.45	12	41
Novichok	security	100	25	0.34	0.08	25	1
DPRK bio-chem weapons	security	100	2	0.56	0.00	2	1
Explosive terror	conventional	100	48	0.68	0.83	48	44
IED	conventional	100	40	0.71	0.87	40	50
Drone attack	conventional	100	22	0.41	0.57	22	13
Mass shooting	conventional	100	53	0.64	0.89	53	50
Vehicle ramming	conventional	100	59	0.46	0.85	59	59
Bomb threat	conventional	100	69	0.64	0.91	69	68
Terror-threat post	conventional	100	53	0.79	0.88	53	58
Public-facility terror	conventional	100	7	0.50	1.00	7	7
Airport bomb report	conventional	100	55	0.49	0.82	55	64
Reactor radiation exposure	conventional	100	3	0.47	0.25	3	5

S3. Event estimates versus the chemical-accident registry

$\hat{N} = V p_E^{\text{corpus}} d/m$ over the overlap window (9 Sep 2021 to 28 Apr 2025).

Substance query	V (window)	p_E^{corpus} (%)	m	Domestic (%)	$\hat{N}$ domestic	Registry incidents
HF leak	244	40	1.25	95	75	12
NH ₃ leak	262	50	1.85	98	69	24
Cl ₂ gas leak	162	36	1.35	100	43	10
H ₂ SO ₄ spill	164	70	1.86	99	61	55
Toluene leak	97	40	1.54	100	25	14
HNO ₃ leak	110	73	2.15	99	37	36

S4. File-to-query assignment

The collection manifest (collection_manifest.csv in the repository) lists every export file with its download-order assignment, its content-based assignment and the token-fit scores against all 30 queries; the nine files whose assignments differ are flagged in the manifest.

S5. Codebook, coder prompt and clustering rule

The codebook (v2), the coder codebook (version 2), the anchor lexicons with spelling variants, the event clustering code and the ten cross-validation seeds (0–9) are provided in the repository. The wrapper text used to present each article to the model coders was not retained.

S6. Query list, occurrence-noun coding, sampling fraction and event share

Query (English)	Block	Query (Korean)	Occurrence noun	<i>n</i>	<i>n/V</i> (%)	<i>p_E</i> (%)
HF leak	Industrial	불산 누출	yes (누출)	100	27.5	40
NH ₃ leak	Industrial	암모니아 누출 사고	yes (누출)	100	21.1	50
Cl ₂ gas leak	Industrial	염소 가스 누출	yes (누출)	100	41.5	35
H ₂ SO ₄ spill	Industrial	황산 유출 사고	yes (유출)	100	39.4	69
Chem. plant explosion	Industrial	화학공장 폭발	yes (폭발)	100	19.0	47
Hazmat release	Industrial	유해화학물질 유출	yes (유출)	67	100.0	9
Toluene leak	Industrial	톨루엔 누출	yes (누출)	100	81.3	40
HNO ₃ leak	Industrial	질산 누출	yes (누출)	100	46.3	73
Toxic-gas evacuation	Industrial	유독가스 누출 대피	yes (누출)	100	69.4	55
Chem. accident response	Industrial	화학사고 대응	no	100	4.5	3
Chemical weapon use	Security	화학무기 사용	yes (사용)	100	9.6	11
Sarin attack	Security	사린가스 공격	yes (공격)	41	100.0	7
Bioterrorism	Security	생물테러	no	100	6.9	1
Anthrax terror	Security	탄저균 테러	no	100	32.4	7
DPRK nuclear test	Security	북한 핵실험	yes (실험)	100	0.4	7
Radiation leak	Security	방사능 유출	yes (유출)	100	3.7	28
Dirty bomb	Security	더티밤	no	100	42.0	4
White-powder report	Security	백색가루 신고	yes (신고)	59	100.0	12
Novichok	Security	신경작용제 노비쵸크	no	100	44.6	25
DPRK bio-chem weapons	Security	북한 생화학무기	no	100	29.8	2
Explosive terror	Conventional	폭발물 테러	yes (폭발)	100	1.4	48
IED	Conventional	사제폭탄	no	100	20.5	40
Drone attack	Conventional	드론 공격	yes (공격)	100	0.2	22
Mass shooting	Conventional	총기 난사	yes (난사)	100	1.2	53
Vehicle ramming	Conventional	차량 돌진 테러	yes (돌진)	100	7.4	59
Bomb threat	Conventional	폭발물 협박	yes (폭발/협박)	100	2.4	69
Terror-threat post	Conventional	테러 예고 글	yes (예고)	100	3.8	53
Public-facility terror	Conventional	다중이용시설 테러	no	100	7.4	7
Airport bomb report	Conventional	공항 폭발물 신고	yes (폭발/신고)	100	33.2	55
Reactor radiation exposure	Conventional	원전 방사선 피폭	yes (피폭)	100	9.9	3

S7. Novichok EVENT articles in the gold set

Twenty-five articles coded EVENT under the “Novichok” query, with the coder’s event place and object. Fourteen, all dated February 2024, report the death in custody of Alexei Navalny, reported as a suspected poisoning; the other eleven report other suspected poisonings, including three from February 2026. None reports a new use of Novichok.

Date	Coded place	Coded object
2022-03-29	–	독극물 중독
2022-03-29	–	독극물
2022-03-30	우크라이나 키이우	독극물
2023-05-22	베를린	독극물 공격
2023-11-29	우크라이나	중금속 독극물 중독
2024-02-16	러시아 야말로네네츠	의문사
2024-02-16	러시아 야말로네네츠	수감 중 사망
2024-02-17	러시아	의문사
2024-02-17	러시아 야말로네네츠	의문사 독살 의혹
2024-02-17	러시아 야말로네네츠	독극물 의혹 사망
2024-02-18	러시아 하르프	의문사
2024-02-19	살레하르트	독살 의혹 사망
2024-02-20	러시아	노비추크 독살
2024-02-21	러시아	암살 의혹
2024-02-21	러시아	의문사
2024-02-21	러시아	독극물 의혹 사망
2024-02-22	러시아	독극물 의혹 사망
2024-02-25	러시아	의문사
2024-02-26	러시아	의문사
2024-05-02	우크라이나	클로로피크린
2024-05-02	우크라이나	화학무기
2024-11-14	베오그라드	의문사
2026-02-15	러시아	독극물 살해
2026-02-15	러시아	신경독
2026-02-17	러시아	독극물

S8. Pairwise Mann–Whitney tests on query-level p_E : asymptotic and exact p

Comparison	Asymptotic p	Exact p
Security vs industrial	0.0046	0.0029
Security vs conventional	0.0112	0.0089
Industrial vs conventional	0.9396	0.9705

S9. Sensitivity to query assignment and query wording

The block contrast does not depend on the queries that a reader might assign differently. Excluding “DPRK nuclear test” (geopolitical rather than hazard coverage), “chemical accident response” (78% institutional), “reactor radiation exposure” (the boundary query in the conventional block) or “drone attack” (foreign war coverage), singly or all four together, leaves the Kruskal–Wallis p at or below 0.01, the security block below both controls at Holm-adjusted $p \leq 0.03$, and the industrial and conventional blocks indistinguishable ($p \geq 0.65$). Block event shares move by at most five points (industrial 43–48%, security

11%, conventional 41–48%). Moving “reactor radiation exposure” into the security block instead of excluding it strengthens every contrast (Kruskal–Wallis $p = 0.0013$; security versus industrial $p = 0.003$, versus conventional $p = 0.002$).

Query wording is a second candidate confound: most industrial and conventional queries contain an occurrence noun (leak, explosion, attack, shooting, threat), whereas half of the security queries name a threat without one (bioterrorism, dirty bomb, Novichok). Coding each query for the presence of an occurrence noun gives, within the security block, event shares of 13% for the five queries with one and 8% for the five without; the five security queries with an occurrence noun still fall below the industrial block ($p = 0.043$) and below the conventional block ($p = 0.086$, five against ten queries). The occurrence-noun coding of each query is given in the supplement. Wording explains part of the within-block spread, and the industrial and conventional queries without an occurrence noun (“chemical accident response”, 3%; “public-facility terror”, 7%) are the low outliers of their blocks, but it does not account for the block contrast. The event share also varies by year within blocks (security 0–27%, conventional 27–51%, industrial 24–56% across the six calendar years), which is the quantity a planner would track.

S10. Duplication factor on human-confirmed security events

The following is the full version of the recomputation summarised in Section 4.7 of the main text.

Recomputing amplification on the human event share, holding the duplication factor at its model-based value from Table 1, gives $A = 22.5$ for the security block, 5.2 for the industrial block and 3.1 for the conventional block; no interval is attached, for the reason given at the end of this section and in Section 4.7 of the main text. These are hybrid quantities, with a human numerator share and a model-based duplication factor, and the duplication factor is the weaker half: $m = 1.12$ was computed over the 95 articles coder A called events in this block, of which 18 survive the human standard, so it describes a set that is three-quarters false positives. The human coders identified events but did not record the date, place and object fields the clustering rule needs. Because the security block is small enough to settle by hand, one of us assigned those three fields to all 22 articles the two human coders agreed were events, working from title and excerpt alone, and passed them through the same clustering function the coders’ labels were passed through, with the same three-day window on the event date. This is a third coding pass by a single analyst who was not blind to the analysis; the 22 assignments and their clusters are deposited so that a reader can check them against the titles. They resolve into 12 events, giving $m = 1.83$ against 1.12 on coder A’s labels. The largest cluster is six articles on one white-powder find in a Daegu flat, which coder A had split, and it carries the result: removing it gives 1.45, while removing any other cluster leaves 1.73–1.91. Three further variants bound the figure. On the 18 articles that are both coder-A and human events it is 1.64; splitting the three-week follow-up of the Daegu find into its own event gives 1.69 and merging the two 2024 chloropicrin reports gives 2.00. Weighting the articles by the inverse of their selection probability, so that the duplication factor rests on the same design as p_E , gives 1.57. Taken together the security duplication factor is between 1.45 and 2.00 on human-confirmed events against 1.12 on coder A’s, and A is correspondingly 29 to 40 rather than 22.5. We do not attach an interval to these: with twelve clusters, one of which supplies a quarter of the articles, a resampling interval would be a statement about

that one cluster, and the same 22 articles enter both m and p_E , so the two quantities are not independent and cannot be combined into one. Correcting the denominator moves the estimate away from the controls, not towards them; 22.5 remains the conservative figure and is the one we carry forward and report in the abstract. The block ordering is unchanged and the gap between the security block and the controls is wider than the model labels showed.

S11. Sensitivity of the duplication factor to the clustering rule

Windows of one, three and seven days give m between 1.73 and 1.77 (industrial), 1.10 and 1.12 (security) and 1.18 and 1.19 (conventional). Dropping place or object as a matching criterion raises m in every block, to 2.2–2.5 in the industrial block, and dropping both raises it to 4.2, 2.0 and 2.7. The block ordering is unchanged under every variant.

S12. Classifier agreement rates with coder A, by block

Sensitivity and false-positive rate for the EVENT class, measured against coder A’s labels in cross-validation (10 seeds $\times$ 5 folds). These are agreement rates with the model annotator, not validity against the human reference standard, and they are not the rates used in the misclassification correction of the main text.

Block	Sensitivity	False-positive rate
Industrial	0.93	0.08
Conventional terror	0.89	0.12
Security (CBRN)	0.43	0.02

Ablation differences not shown in the main text: word n -grams are worse than character n -grams by 0.043 macro- F_1 (95% CI 0.038–0.047, corrected $p = 1.5 \times 10^{-4}$) and title-only input by 0.066 (0.060–0.071, $p = 4.5 \times 10^{-5}$). The archive-taxonomy feature lowers EVENT- F_1 by 0.005 ($p = 0.35$) alongside the macro- F_1 difference reported in the main text. Learning-curve means are 0.592, 0.666, 0.699 and 0.710 at the 25, 50, 75 and 100% training fractions.

S13. Treatment of the zero-count cell

Block event shares are post-stratified onto six cells formed by block and coder A’s EVENT-versus-rest decision. The table gives, for each cell, the number of agreed human events, the number of agreed articles drawn, the cell’s size in the gold set, and a 95% Jeffreys interval on the cell’s own share.

Cell	Human events	n agreed	N gold	Jeffreys 95%
Industrial, coder A EVENT	23	32	418	0.549–0.851
Industrial, coder A non-EVENT	2	42	549	0.010–0.144
Security, coder A EVENT	18	79	95	0.146–0.329
Security, coder A non-EVENT	4	140	805	0.010–0.066
Conventional, coder A EVENT	26	28	409	0.790–0.985
Conventional, coder A non-EVENT	0	39	591	0.000–0.062

The conventional non-EVENT cell holds no human event. Any resampling scheme that draws within cells returns that cell's share as zero in every resample, so the cell contributes no variance and the block share cannot exceed coder A's value; the coincidence between the two in Table 6 of the main text follows from the zero count rather than from agreement. Carrying the Jeffreys limits for each cell through the post-stratification weights gives block envelopes of 24.3–45.0% (industrial), 2.4–9.4% (security) and 32.3–43.9% (conventional). These take both cells' limits at once and are therefore conservative envelopes rather than 95% intervals, and they resample nothing: the security envelope is narrower than that block's query-clustered interval for that reason, not because the zero-cell treatment tightened anything. The conventional upper limit is then 43.9% rather than 40.9%, and the cap is removed. These figures assume independence within cells and are therefore not a substitute for the query-clustered intervals reported in the main text, which match the unit of analysis; they are given to show that the cap is an artefact of a zero count and that no conclusion rests on it. The same zero count makes coder A's conventional sensitivity of 1.00 a point estimate without an upper constraint; taking the Jeffreys upper limit on the missed-event cell gives a lower bound of 0.91.

S14. Recall of the registry substance matching

The strict recode of Section 4.5 removes false matches. This section looks for the opposite error. For each of the six substances we searched the incident description and cause fields of all 366 records in the overlap window under an expanded set of names, including anhydrous and waste forms and salt names, and adjudicated every record that the substance-field rule had not already matched.

Substance	Date	Record	Verdict
NH ₃	2025-03-10	Substance field nitric acid (60%); ammonia water named only as the delivery being made when the nitric acid fell	Not a release
H ₂ SO ₄	2022-12-14	Substance field sodium hydroxide; sulfuric acid plant damaged by the explosion but not released	Not a release
Toluene	2023-01-26	RAW C9+C stream: solvent naphtha, indene, vinyltoluene, styrene. Vinyltoluene is methylstyrene	Different compound
HNO ₃	2025-02-27	Ammonium nitrate sacks fell from a truck	Salt, excluded by the strict rule
HF	2024-08-16	Substance field ammonium bifluoride; description records HF residue in a pipe under replacement, with an injury	Miss

One of the five candidates is a release of the substance the rule was looking for. Adding it gives recall 0.92 for hydrogen fluoride and 1.00 for the other five substances, 142 of 143 releases. The log-log slope moves from -0.036 ($p = 0.90$) to -0.023 ($p = 0.94$), and the hypothesis of a slope of 1 is still rejected ($t(4) = 3.6$, $p = 0.023$). The audit code and its output

are deposited as `analyze_22_registry_recall.py` and `registry_recall.json`.